

Item display position interpolation not working

MC - 259915 Commands

Status	Resolved / Fixed
Affected	23w06a, 1.19.4 Pre-release 3, 1.19.4 Release Candidate 2, 1.19.4, 1.20 Pre-release 1, 1.20.1
Fixed in	23w07a, 23w31a
Reported	2023-02-10

STEPS TO REPRODUCE

1. Launch Minecraft and create a Creative world with default settings.
2. Create an item display entity and configure it to display an item.
3. Set up command blocks to teleport the item display between different positions relative to the player.
4. Activate the command blocks so the item display repeatedly changes position and rotation.
5. Observe that the item display moves abruptly between positions instead of smoothly interpolating its position and rotation.

OBSERVED BEHAVIOUR

Actual behavior

When an Item Display entity's position or rotation is updated, the client renders the change as an immediate snap to the new transform. It does not interpolate between the previous and new position or between the previous and new yaw/pitch, so repeatedly teleporting or moving the entity appears visibly jarring rather than smooth.

This occurs when the entity is moved using command blocks as well as through Fabric-based code, and is reproducible in both single-player and a local server. Rapid or frame-timed updates can also produce unstable visual results when pausing and unpausing the game.

The behavior is consistent with the entity not receiving the position interpolation normally provided by some ticking entities. Applying movement through TRSR transformation does not provide a reliable workaround, as the resulting motion is unstable and resource-intensive. Mounting the display on another entity is also not currently a reliable workaround because of the related mounting issue described in MC-259816.

EXPECTED BEHAVIOUR

When an item display's position or rotation changes, it should smoothly interpolate between its previous and new transformation rather than snapping or appearing to teleport abruptly. This interpolation should remain stable during normal movement and updates, allowing item displays to be used reliably as moving objects and providing behavior comparable to other rendered entities with smooth positional transitions.

ENVIRONMENT

Minecraft Java Edition Versions: 23w06a, 1.19.4 Pre-release 3, 1.19.4 Release Candidate 2, 1.19.4, 1.20 Pre-release 1, 1.20.1 (includes snapshots)

ORIGINAL DESCRIPTION

Want to start off this report by tipping my hats to the devs and expressing gratitude at how incredible of a feature this is for developers and will open up an unspeakable amount of new opportunities, thank you for giving this to the community.

While messing with the new feature after it came out, came to a disappointing realization that there seems to be no position interpolation (showcased in the video: https://youtu.be/H-_zwxMdP_U). Based on the nature of this feature, I'm led to believe the lack of this would be a bug as it makes teleporting it quite jarring and not usable as a moving thing (which opens up so many more opportunities).

Some other notes that may be relevant as to this bug:

- because of the non-ticking nature of this entity, it doesn't have the built-in position interpolation that living entity has, though the usage properties of it make position interpolation very relevant.
- similar behavior can be seen with area effect clouds even though it is a ticking entity

- marker entity gets a pass to be non-ticking because it is completely server side and does not render on client side
- position interpolation can be mimicked with TRSR transformation, but the results are unstable and resource intensive. This was a tested alternative to this bug before reporting, but between this and the mounting not working we came to this conclusion. <https://youtu.be/qD5JjeXNGvI>

Tests were ran both through command blocks to achieve the teleporting to me, as well as verified on the latest versions of fabric as we're more comfortable doing that kind of development over the command system, but can verify it on both vanilla and fabric, as well as singleplayer and on a local server. This video shows a fabric version of the testing (compared to vanilla on the others) where you can see unstable results when pausing/unpausing the games and using frame perfect packets to move the entity: https://youtu.be/I_PKHfFJ_BY

Was doing some searching of the issues and found this issue

MC-259816

Resolved

which addresses it in a similar way, at this time you can't reliably mount these display items on other entities without it breaking, allowing for NO way to s

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-259915>